

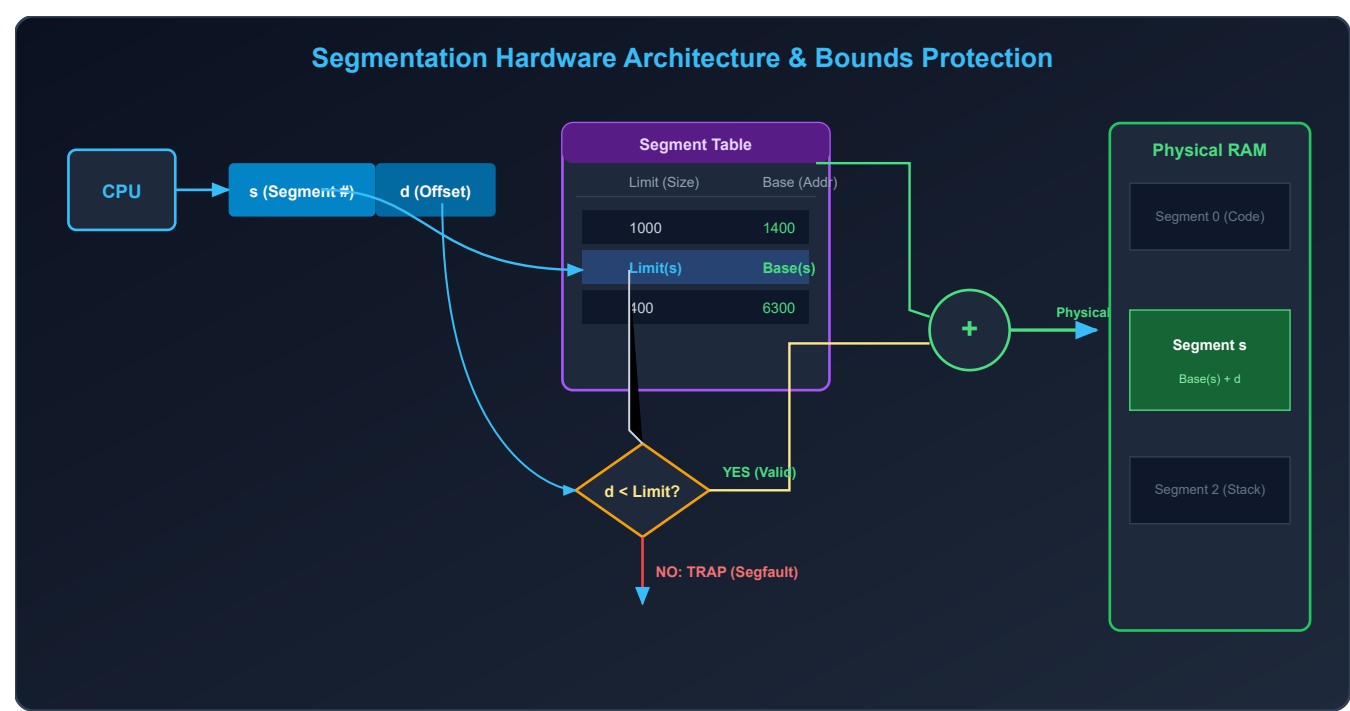
Module 07: Segmentation and Paged Segmentation

1. Programmer View vs OS View

Segmentation memory ko logical units (Code, Stack, Heap, Data) me organize karta hai. Address $\langle s, d \rangle$ hota hai jaha s = Segment number aur d = Offset.

2. Hardware Bounds Protection

Segment Table me do fields hote hain: **Base** (Physical start) aur **Limit** (Length). Agar $d \geq \text{Limit}$, hardware MMU ek **Trap (Segmentation Fault)** generate karta hai. Agar $d < \text{Limit}$, tab Physical Address = Base + d.



3. Paging vs Segmentation Comparison

Parameter	Paging	Segmentation
Block Size	Fixed (4 KB)	Variable
Decided by	Hardware / OS	Programmer / Compiler
Internal Fragmentation	Yes (in last page)	No (Zero)
External Fragmentation	No (Zero)	Yes
Address Calculation	Bit Concatenation	Base + Offset Addition

4. Solved Examples

- $\langle 0, 430 \rangle$: $430 < 600$ (Valid) $\rightarrow$ Base $219 + 430 = \mathbf{649}$.
- $\langle 2, 500 \rangle$: $500 \geq 100$ (Limit Exceeded) $\rightarrow$ **Trap: Segfault.**
- $\langle 3, 400 \rangle$: $400 < 580$ (Valid) $\rightarrow$ Base $1327 + 400 = \mathbf{1727}$.